

CASE 38

MULTIPLE SCLEROSIS

An autoimmune attack on the central nervous system.

As we saw in rheumatoid arthritis (Case 35), autoimmune disease can be caused by activated effector T cells specific for self peptides. When T cells recognize self-peptide:MHC complexes and become activated, they can cause local inflammation by activating macrophages and other cells with consequent tissue damage. Another example of a T cell-mediated autoimmune disease is the neurologic disease multiple sclerosis (MS). However, unlike the certain involvement of autoantibodies in some types of autoimmune diseases (see Case 40), it has been difficult to prove the involvement of T cells in MS, because T cells do not cross the placenta into the fetus and experimental T-cell transfer is not allowed in humans.

More than seventy years ago, an experimental model of MS was established in mice, in which the injection of myelin in adjuvant caused the development of neurologic symptoms similar to those of MS. This disease is called experimental autoimmune encephalomyelitis (EAE). The antigens in myelin that can induce EAE are myelin basic protein (MBP), proteolipid protein (PLP), and myelin oligodendrocyte glycoprotein (MOG). The disease can be transferred to syngeneic animals by cloned antigen-specific T-cell lines derived from animals with EAE (Fig. 38.1). When the recipient animals are immunized with MBP, for example, they develop active disease. T cells specific for MBP, PLP, and MOG have been found in the blood and cerebrospinal fluid (CSF) of patients with MS.

TOPICS BEARING ON THIS CASE:

Inflammatory reactions

Interactions of co-stimulatory molecules with their receptors

The development of tolerance to self antigens

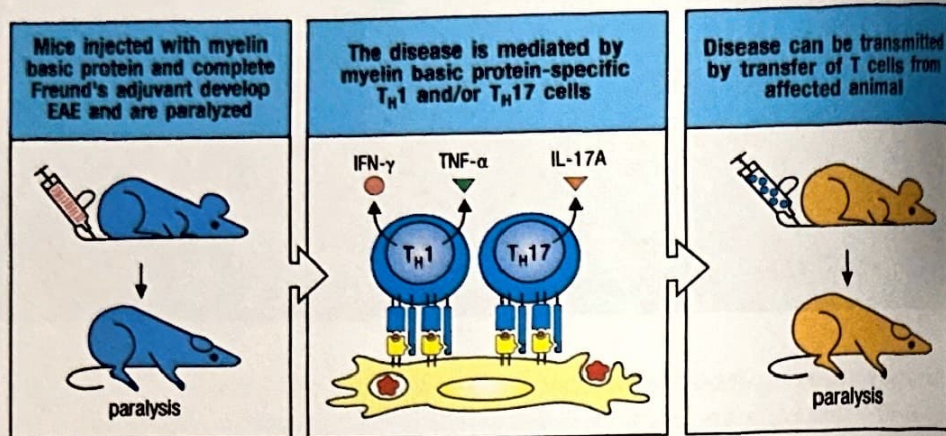
Activation of self-reactive T cells

Immunologically privileged sites

Experimental autoimmune encephalomyelitis

Induction of oral tolerance

Fig. 38.1 T cells specific for myelin basic protein mediate inflammation of the brain in experimental autoimmune encephalomyelitis (EAE). This disease is produced in experimental animals by injecting them with isolated spinal cord homogenized in complete Freund's adjuvant. EAE is due to an inflammatory reaction in the brain that causes a progressive paralysis affecting first the tail and hindlimbs before progressing to forelimb paralysis and eventual death. One of the autoantigens identified in the spinal cord homogenate is myelin basic protein (MBP). Immunization with MBP alone in complete Freund's adjuvant can also cause these disease symptoms. Inflammation of the brain and paralysis are mediated by T_H1 and/or T_H17 cells specific for MBP. Cloned MBP-specific T_H1 and/or T_H17 cells can transfer symptoms of EAE to naive recipients provided that the recipients carry the correct MHC allele. In this system it has therefore proved possible to identify the peptide:MHC complex recognized by the T_H clones that transfer disease.



The case of Vivie Warren: an oboist who has difficulty reading a musical score.

Mrs Vivie Warren, a 29-year-old professional oboe player, was in good health until one morning she noticed a loss of vision in her left eye. Her physician referred her to a neurologist, who found that her eye movement was normal and not accompanied by any pain. The visual acuity in Vivie's left eye was 20/100 and in her right eye 20/200. Her retina was normal, and a detailed neurologic examination also proved normal. The neurologist diagnosed optic neuritis (inflammation of the optic nerve). Her family history, however, revealed that her mother had severe MS and was permanently disabled, and a magnetic resonance imaging (MRI) brain scan was ordered.

Vivie was given a 5-day course of intravenous corticosteroids, and her vision returned to normal over the next 3 weeks. The MRI scan revealed multiple lesions in the white matter of the brain under the cortex and around the ventricles (Fig. 38.2). Intravenous injection of gadolinium, a contrast agent that leaks from blood vessels in recently inflamed tissue, showed that some of the brain lesions were probably of recent origin (Fig. 38.3). The neurologist told Vivie that she had a high probability of developing MS and advised her to return for frequent neurologic examination.

Vivie remained well for a further 3 years and then developed weakness of the muscles on the left side of her face that were innervated by the seventh cranial nerve. A repeat MRI scan with gadolinium enhancement showed new lesions in the left middle cerebellar peduncle and in the pons. CSF was obtained by lumbar puncture. It contained 28 mg dl⁻¹ protein (normal) and 8 lymphocytes ml⁻¹ (normal 0–3 ml⁻¹). At this point a firm diagnosis of MS was made. Despite the normal level of protein in the CSF, the IgG content was raised. On electrophoresis, discrete bands of IgG were observed, indicating clonal expansion of restricted B-cell populations in the central nervous system (CNS). Another 5-day course of corticosteroids was administered intravenously, and Vivie's symptoms improved. Weekly intramuscular injections of interferon (IFN)- β were started to prevent progression of the disease.

Fig. 38.2 Three-dimensional magnetic resonance images of the brain at three different time points in the course of MS. Left, early; center, during acute exacerbation; right, after therapy. The technique used causes fluid to appear white. The lateral ventricles in the middle of the brain scan and the sulci of the cerebral cortex around the edge appear white as a result of normal cerebrospinal fluid. The white spots, which are due to edema fluid and decreased myelin, are MS lesions.





Fig. 38.3 Three computer-generated images of MRI scans of the brain shown in Fig. 38.2 at the same time points. Several levels of MRI scans have been integrated by computer. The lateral and middle ventricles appear in purple. Old MS lesions appear yellow-green. A new, gadolinium-enhanced, lesion appears orange. On the right, this lesion has diminished in size after immunosuppressive therapy.

Vivie did well for 3 more years, after which she developed a weakness in her left leg and left hand. Her speech became slurred. She developed nystagmus (rapid uncontrolled horizontal jerking eye movements when attempting to fix the gaze on something) and ataxia (wide-based staggering gait). Vivie was given another course of corticosteroids, after which her symptoms improved, but 8 months later they recurred. The injections of IFN- β were stopped and she was put on high doses of cyclophosphamide and corticosteroids at monthly intervals. After 3 months of this therapy, the cyclophosphamide and corticosteroid injections were gradually reduced to every 12 weeks. Her neurological examination became normal and no new lesions were observed on gadolinium-enhanced MRI.

Multiple sclerosis.

Multiple sclerosis (MS) was first described by the great French neurologist Jean-Martin Charcot in the 1860s. It was noted at autopsy that patients who died of this disease had multiple hard (sclerotic) plaques scattered throughout the white matter of the CNS. The disease is 10 times more frequent in women than in men and is associated with HLA-DR2. Those affected have a variety of nervous symptoms, such as urinary incontinence, blindness, ataxia, muscle weakness, and paralysis of limbs. The plaques characteristic of the disease show dissolution of myelin along with infiltrates of lymphocytes and macrophages, particularly along blood vessels. The inflammatory exudate causes increased vascular permeability.

The CNS is a relatively immunologically privileged site from which antigens do not normally reach the lymphoid tissues, and so there is no negative selection of T cells that have the potential to react against CNS antigens. In MS, an unknown injurious event is presumed to provoke the release of CNS antigens and their presentation to lymphocytes in the peripheral lymphoid organs. This results in the expansion of clones of autoreactive T cells and their differentiation into T_H1 cells, which home to the CNS and initiate inflammation. These T_H1 cells can be readily identified in the CNS of patients with MS.

Lymphocytes and other blood cells do not normally cross the blood-brain barrier. If tissue becomes inflamed, however, activated $CD4^+$ T cells autoreactive for a brain antigen and expressing $\alpha_4\beta_1$ integrin, which binds vascular cell adhesion molecules (VCAM) on the surface of activated venule endothelium, can migrate out of the blood into the brain. There they reencounter their specific autoantigen presented by MHC class II molecules on microglial cells and produce pro-inflammatory cytokines such as IFN- γ (Fig. 38.4). Microglia are phagocytic macrophage-like cells of the innate immune system resident in the CNS and, like macrophages, can act as antigen-presenting cells. Inflammation causes increased vascular permeability, and the site becomes heavily infiltrated by activated macrophages and T_H1 cells, which produce pro-inflammatory cytokines that exacerbate the inflammation, resulting in the further recruitment of T cells, B cells, macrophages, and dendritic cells to the site of the lesion. Autoreactive B cells produce autoantibodies

Relapse; give weekly IFN- β .

Relapse; aggressive immunosuppressive therapy started.

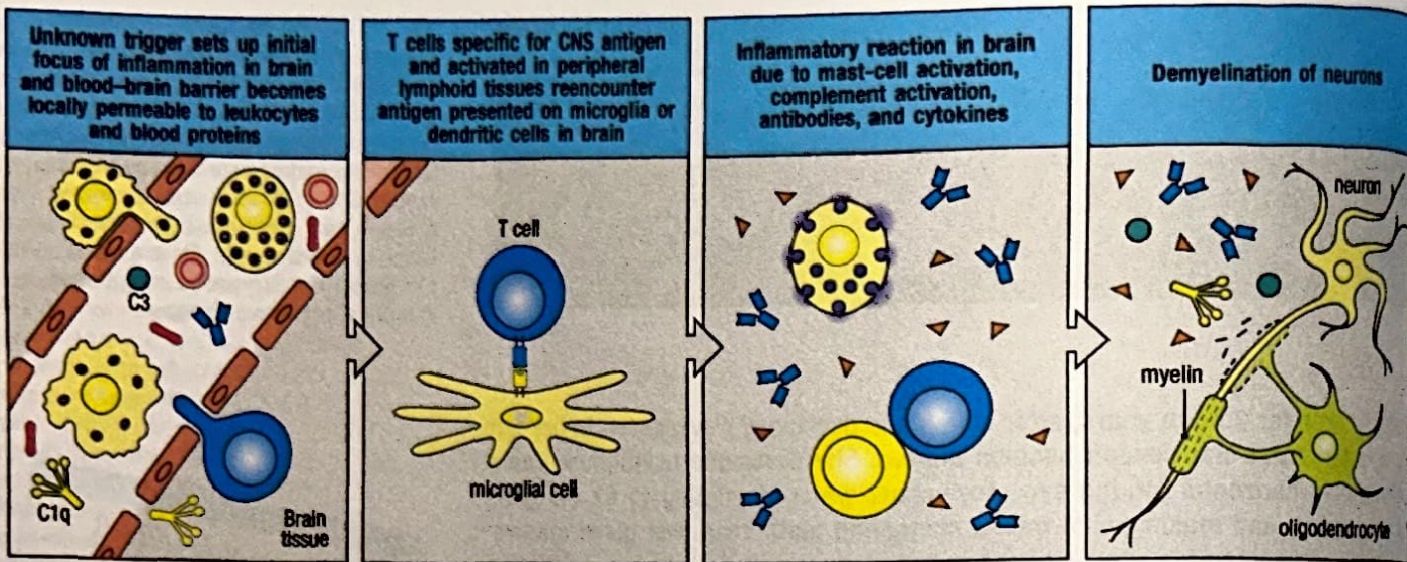


Fig. 38.4 The pathogenesis of multiple sclerosis. At sites of inflammation, activated T cells autoreactive for brain antigens can cross the blood-brain barrier and enter the brain, where they reencounter their antigens on microglial cells and secrete cytokines such as IFN- γ and IL-17A. The production of T-cell and macrophage cytokines exacerbates the inflammation and induces

a further influx of blood cells (including macrophages, dendritic cells, and B cells) and blood proteins (such as complement) into the affected site. Mast cells also become activated. The individual roles of these components in demyelination and loss of neuronal function are still not well understood.

against myelin antigens with help from T cells. Activated mast cells release histamine, contributing to the inflammation. In some way that is not yet fully understood, these combined activities lead to demyelination and interference with neuronal function.

Mice deficient in IFN- γ are not protected from the development of EAE, suggesting that additional T_H cells may drive CNS inflammation. T_H17 cells are a helper T-cell population that can be induced from memory T cells by a combination of cytokines including IL-6, IL-21, and transforming growth factor- β (TGF- β) and are sustained by IL-23. Although T_H17 cells do not seem to mediate inflammation in all contexts, studies in mouse EAE and human MS reveal a pathologic role for T_H17 cells and their associated cytokine, IL-17A. Adoptive transfer of MOG-specific T_H1 and T_H17 cells differentiated *in vitro* induces EAE with a distinct histologic appearance, indicating that several effector T-cell types drive autoimmunity in MS. Autoreactive T_H1 cells with specificity for MS-associated myelin antigens can be found in healthy patients without MS. Investigators searching for the suppressive mechanisms that inhibit potentially autoreactive T cells in healthy individuals have focused on regulatory T cells (T_{reg} cells), a subset of naturally occurring $CD4^+CD25^+$ T cells that promote peripheral tolerance and inhibit autoimmunity in multiple organs. Although patients with MS have normal numbers of T_{reg} cells, the cells have a decreased ability to suppress autoreactive T cells *in vitro*. Future research will seek to further investigate the regulation of T_H -cell subtypes and the interplay of autoreactive and suppressive T cells in MS.

In mice, feeding with MBP before immunization with MBP in adjuvant protects against the development of EAE. MBP-specific T cells can be identified in the brain of protected mice, but they secrete TGF- β rather than IFN- γ and fail to initiate an inflammatory reaction. Furthermore, adoptive transfer of T cells from mice fed MBP can confer protection against EAE. Therapeutic attempts to treat patients with MS by oral MBP have not been successful, however. This suggests that therapeutic interventions that induce oral tolerance may not be effective in already established disease.

Questions.

- 1 Oligoclonal immunoglobulins were found in Mrs Warren's central nervous system. How do you explain this?
- 2 Mrs Warren was treated with corticosteroids, cyclophosphamide, and IFN- β . What was the aim of this therapy?
- 3 An attempt has been made to treat MS patients with IFN- γ . Can you predict what the outcome was and why?
- 4 What is the rationale behind feeding MBP to mice to prevent EAE?
- 5 Can you predict whether EAE can be induced in CD28 knockout mice?